

S.T.A.R. Claim/Evidence Registry

Version 0.2

Scientific Claim, Evidence, Provenance, and Validation Control System

1. Purpose

The S.T.A.R. Claim/Evidence Registry records every substantive mathematical, physical, astronomical, computational, statistical, and methodological claim made within the S.T.A.R. research program.

The registry exists to prevent exploratory results, historical interpretations, mathematical derivations, computational observations, and empirically validated results from being treated as equivalent forms of evidence.

The registry therefore separates:

1. **what is mathematically derived;**
2. **what has been computationally reproduced;**
3. **what has been empirically observed;**
4. **what is a model or phenomenological construction;**
5. **what is hypothesized;**
6. **what is conjectured;**
7. **what constitutes a testable prediction;**
8. **what remains speculative;**
9. **what has failed, become unstable, or remains unresolved.**

The registry is subordinate to the S.T.A.R. Research Charter and is the authoritative index connecting claims to:

[
 \boxed{
 \text{Claim}
 \rightarrow
 \text{Evidence}
 \rightarrow
 \text{Data}
 \rightarrow
 \text{Code}

```
\rightarrow
\text{Experiment}
\rightarrow
\text{Result}
\rightarrow
\text{Replication}
}
]
```

No historical result becomes established evidence merely because it appears in an earlier S.T.A.R. document.

2. Governing Scientific Principle

The registry adopts the following governing principle:

Do not ask whether the theory can be made to fit the universe. Ask whether a fixed mathematical construction predicts structure that was not used to construct it.

Consequently, a claim gains evidentiary status only through a traceable chain of derivation, computation, controlled experiment, independent validation, or replication appropriate to the type of claim.

3. Epistemic Classification System

Every claim shall receive at least one primary epistemic classification.

D — Mathematically Derived

A result follows from explicitly stated mathematical assumptions, definitions, axioms, or previously established results.

Examples:

- algebraic identities;
- proven convergence under stated assumptions;
- Euler–Lagrange equations derived from an explicitly defined action;

- differential/geometric identities.

A D classification does **not** imply that the assumptions describe nature.

E — Empirically Observed

A property measured or directly calculated from an astronomical or experimental dataset.

Examples:

- an observed correlation;
- a measured distribution;
- a reproducible dataset statistic;
- an observed residual pattern.

Evidence must identify the population, dataset, processing procedure, and uncertainty.

M — Model / Phenomenological Construction

A mathematical or computational construction introduced to represent a physical or astronomical system but not yet established as fundamental.

Examples:

[
 $L_{\rm cosmo}(s)$
]

or a proposed arithmetic-to-geometric mapping.

M status is not a claim that the construction is physically correct.

P — Postulate

A foundational assumption deliberately introduced by the theory.

Postulates are not empirical evidence.

H — Hypothesis

A proposed relationship whose truth remains to be determined experimentally or mathematically.

Examples:

Arithmetic rank may correspond to cosmic-web complexity.

C — Conjecture

A stronger mathematical or theoretical proposition proposed for future proof or falsification.

T — Testable Prediction

A claim that produces a measurable consequence that can distinguish the proposed model from an appropriate null or competing model.

A T classification should identify:

- observable;
 - dataset;
 - expected relation;
 - null;
 - acceptance/failure criterion.
-

S — Speculative Interpretation

An interpretation extending beyond presently demonstrated mathematics or empirical evidence.

S claims must not be presented as established conclusions.

4. Evidence Provenance Classification

In addition to epistemic classification, every evidentiary item receives a provenance classification.

THEORETICAL

Derived from mathematical reasoning.

SIMULATION

Produced through controlled synthetic data or numerical simulation.

PILOT OBSERVATION

Observed in real data but not yet sufficiently controlled or independently replicated.

HISTORICAL MODEL

Generated by an earlier S.T.A.R. model, script, or analysis.

CONTROLLED EXPERIMENT

Produced by a registered experiment with locked methodology and appropriate controls.

INDEPENDENT REPLICATION

Repeated independently using data, code, or methodology not used to establish the original result.

EXPLORATORY

Useful for discovery but not sufficient to establish evidence.

NEGATIVE / NULL

A null, failed prediction, instability, or result inconsistent with the proposed relationship.

UNVERIFIED

A result or statement for which the supporting computation, derivation, or evidence has not yet been independently checked.

5. Evidence Status

Each claim shall receive one current evidence status.

Status	Meaning
PROPOSED	Claim entered into registry but not yet tested
DERIVED	Mathematical derivation completed
COMPUTATIONALLY REPRODUCED	Same computational result reproduced
PILOT SUPPORTED	Initial empirical support exists
CONTROLLED	Tested against registered controls/nulls
REPLICATED	Independently reproduced
PREDICTIVE	Demonstrated out-of-sample predictive correspondence
PHYSICALLY CONSTRAINED	Compared against established physical/observational constraints
FALSIFIED	Registered test contradicts claim under its stated assumptions
INCONCLUSIVE	Evidence insufficient to decide
SUPERSEDED	Replaced by a better-defined formulation
HISTORICAL ONLY	Retained for provenance but no longer treated as active evidence

6. Evidence Quality and Reproducibility

Each claim shall additionally record:

Reproducibility

- NOT ATTEMPTED
- PARTIAL
- REPRODUCED
- INDEPENDENTLY REPRODUCED

Leakage / Contamination Status

- UNKNOWN
- POTENTIAL LEAKAGE
- CONFIRMED LEAKAGE
- NO LEAKAGE FOUND
- RECONSTRUCTED AFTER LEAKAGE CONTROL

This field is mandatory for machine-learning and statistical claims.

Derived quantities must not be calculated using information from held-out data unless that operation is explicitly part of the registered methodology.

7. Required Registry Fields

Every registry entry should contain:

Field	Description
Claim ID	Permanent unique identifier
Statement	Exact claim
Classification	D/E/M/P/H/C/T/S
Category	Mathematical, cosmological, SFR, topology, RTCH, data, etc.
Source Artifact	Document, paper, notebook, script, or archive
Source Version	Version/date where available
Mathematical Basis	Equations or derivation supporting claim
Dataset ID	Dataset supporting empirical component
Code Artifact	Script/notebook used
Experiment ID	Registered experiment testing claim
Prediction	Measurable consequence
Null Model	Required null/alternative
Parameters	Locked parameters or parameter family

Evidence Provenance	Theoretical/simulation/pilot/etc.
Evidence Status	Current status
Reproducibility	Reproduction state
Leakage Status	Statistical contamination state
Dependencies	Claims/results required
Known Limitations	Current limitations
Negative Evidence	Known contradictory/unstable results
Last Tested	Date
Replication Status	Independent replication state
Current Interpretation	Current scientific interpretation

8. Foundational S.T.A.R. Claims

CORE-001 — Arithmetic–Cosmic Structural Correspondence

Statement

Arithmetic invariants may be mapped into geometric or topological structures in a way that produces statistically nontrivial and reproducible correspondence with independent astronomical data.

Classification: H / T

Category: Foundational / Cosmic Structure

Current Status: PROPOSED

Evidence Provenance: EXPLORATORY

Required Test

```
[
E
\rightarrow I(E)
\rightarrow \Phi(E)
\rightarrow \mathrm{H}_{\mathrm{arith}}
\rightarrow D_{\mathrm{arith}}
\rightarrow D_{\mathrm{cosmic}}
]
```

followed by statistical comparison and independent validation.

Required Nulls

- randomized arithmetic assignments;
- permutation controls;
- alternative arithmetic invariant sets;
- alternative mappings;
- survey/random spatial controls.

Interpretation

This is the canonical S.T.A.R. research question rather than an established physical conclusion.

9. Arithmetic Rank and Cosmic Structure

RANK-001 — Rank/Environment Correspondence

Statement

Elliptic-curve rank may contain information corresponding to cosmic-web environment or structural complexity.

Classification: H / T

Category: Arithmetic → Cosmic Structure

Current Status: PROPOSED

Evidence Provenance: EXPLORATORY

Required Comparison

Rank must be compared against:

[
r, $\quad$
 $\operatorname{Reg}(E)$, $\quad$
 $N(E)$, $\quad$
 $|\Delta(E)|$, $\quad$
 $T(E)$
]

and combinations thereof.

Required Controls

Arithmetic permutation and non-rank invariant controls.

RANK-002 — Rank/Topological Complexity

Statement

Arithmetic rank may be associated with measurable differences in persistent-homology or geometric descriptions of cosmic structure.

Classification: H / T

Current Status: PROPOSED

Required Observables

[
 β_0 , $\quad \beta_1$, $\quad \beta_2$
]

and persistence-diagram/landscape/image distances.

Important Constraint

(β_1) shall not be interpreted directly as the number of filaments.

Filament identification shall rely primarily on geometric, skeleton, or graph methods, with persistent homology used as supporting topological information.

RANK-003 — Rank-Stratified Cosmic Structure

Statement

Rank-stratified arithmetic samples may produce statistically distinguishable cosmic environments.

Classification: H / T

Status: PROPOSED

Experiment: Rank/environment and topology ladder.

10. Arithmetic Mapping Claims

MAP-001 — Arithmetic Projection

Statement

A mapping of the form

$$\left[\begin{array}{l} \Phi(E) = \\ \left(\begin{array}{l} \log |\Delta(E)|, \\ \log N(E), \\ \alpha_r(E) \end{array} \right) \end{array} \right]$$

can serve as a candidate projection from arithmetic invariant space into geometric coordinate space.

Classification: M

Evidence Status: EXPLORATORY

Interpretation

The mapping is a model construction, not a demonstrated physical law.

MAP-002 — Extended Projection

Historical S.T.A.R. work considered

$(\phi_E, \theta_E, z_E, s_E)$
]

with coordinates derived from discriminant, conductor, rank, and regulator-related quantities.

Classification: M

Status: HISTORICAL / EXPLORATORY

MAP-003 — Rank Elevation

Historical work used

[
 $z=200r.$
]

Current Classification: M / H

Status: HISTORICAL MODEL

Current Interpretation

The value

[
 $\lambda_r=200$
]

is a candidate scaling parameter:

[
 $h_r=\lambda_r r.$
]

It shall not be treated as a uniquely justified constant.

Alternative values and mappings must be tested.

MAP-004 — Mapping Non-Bijectivity

Statement

A valid arithmetic-to-cosmic mapping need not be bijective; equivalence classes or many-to-one projections may be mathematically and experimentally permissible.

Classification: M / H

Status: ACTIVE METHODOLOGICAL PRINCIPLE

This is consistent with the alternative-mapping framework.

MAP-005 — Alternative Mapping Families

Candidate mappings should be compared against alternatives including the MCJ, PTD, FT, and other registered constructions.

Classification: M / T

Status: PROPOSED EXPERIMENTAL REQUIREMENT

No mapping should receive privileged status merely because it was historically developed first.

11. Fibonacci / Lucas / Arithmetic Generator Claims

ARITH-001

Statement

Fibonacci and Lucas sequences may provide useful arithmetic generator families for constructing candidate mappings.

Classification: M / H

Status: EXPLORATORY

Required Controls

Compare against:

- prime-based constructions;
- randomized recurrence sequences;
- alternative integer sequences;
- parameter-matched controls.

The appearance of a correspondence using Fibonacci/Lucas sequences alone shall not establish physical significance.

12. Topological Evidence Claims

TOPO-001 — Persistent Homology Correspondence

Statement

Persistent homology may detect statistically meaningful correspondence between arithmetic-derived structures and observed cosmic-web topology.

Classification: H / T

Status: PROPOSED

Required Metrics

- persistence diagrams;
- persistence landscapes/images;
- Wasserstein distance;
- bottleneck distance;
- null distributions.

Historical thresholds such as

[
 $W_2 < 10^{-2}$
]

shall be treated as model-dependent exploratory thresholds until calibrated against appropriate nulls, resolution, normalization, noise, and alternative mappings.

TOPO-002 — Void Structure

Statement

Arithmetic-derived structures may contain information corresponding to cosmic void topology.

Classification: H / T

Status: PROPOSED

Relevant Topology

[
 $\backslash\beta_2$
]

and geometric void measurements.

TOPO-003 — Connected Components

Statement

Arithmetic-derived structures may correspond to connected cosmic structures.

Classification: H / T

Relevant Quantity

[
 $\backslash\beta_0.$
]

13. ($K_{\rm theory}$) / Reciprocal-Normalization Claims

MATH-K001 — Log-Normal Reciprocal Normalization

Statement

For a sample of masses following the stated log-normal assumptions, the reciprocal normalization

$$\frac{\sum_i (M_i/M_0)}{\sum_i (M_0/M_i)}$$

approaches unity under the stated assumptions as sample size increases.

Classification: D / H

Evidence Provenance: THEORETICAL / SIMULATION

Current Status: RECONSTRUCT

Required Reconstruction

The mathematical result must be separated from the historical interpretation involving BSD or cosmological structure.

For

$$\begin{aligned} &[\\ &\log M_i \sim N(\mu, \sigma^2), \\ &\quad \quad \quad \\ &M_0 = e^\mu, \\ &] \end{aligned}$$

the relevant expectation symmetry provides the mathematical basis for the limiting result.

Required Controls

- log-normal samples;
- skewed distributions;
- heavy-tailed distributions;
- mixtures;
- empirical galaxy-mass distributions.

MATH-K002 — Empirical (K) Consistency

Historical simulations reported values near unity under specified log-normal conditions.

Classification: E / SIMULATION

Status: HISTORICAL MODEL RESULT

Historical values such as

$$\left[K_{\text{approx}0.998108} \right]$$

shall be retained as provenance rather than interpreted as cosmological validation.

MATH-K003 — Small-Sample Instability

Historical analyses produced substantially different K values under different samples, including small samples.

Classification: E

Status: NEGATIVE / DIAGNOSTIC

This evidence demonstrates that sampling, filtering, distributional assumptions, and dataset definition materially affect (K).

14. ($L_{\text{cosmo}}(s)$)

COSMO-L001 — Phenomenological (L_{cosmo})

Statement

$$\sum_{i=1}^N \left(\frac{M_0}{M_i} \right)^s$$

is a candidate phenomenological statistic for representing galaxy-population structure as a function of mass and redshift.

Classification: M / H

Status: EXPLORATORY

Evidence Provenance: HISTORICAL MODEL

Required Baseline

Compare against ordinary:

- stellar mass;
- gas mass;
- redshift;
- morphology;
- metallicity;
- attenuation;
- age.

The statistic must demonstrate incremental information beyond these variables before receiving stronger evidentiary status.

COSMO-L002 — ($L_{\rm cosmo}$) and SFR

Statement

($L_{\rm cosmo}(s)$) may contain information relevant to predicting star-formation rate.

Classification: H / T

Status: PROPOSED

Critical Requirement

All derived quantities must be constructed inside the appropriate training folds to prevent leakage.

15. Morphology/Redshift Composite Feature

COSMO-R001

Historical S.T.A.R. scripts used a quantity called `cosmo_rank`, constructed from morphology and redshift.

Current Interpretation

This quantity is not elliptic-curve rank.

Classification: M

Status: HISTORICAL FEATURE ENGINEERING

For future work it should be renamed, for example:

```
[  
R_{\rm cosmo}^{\rm feat}  
]
```

or another unambiguous identifier.

It shall not be used as evidence for an elliptic-curve rank relationship.

16. Star Formation Rate Claims

SFR-A001 — Arithmetic Incremental SFR Information

Statement

Arithmetic-derived features may provide out-of-sample information about SFR beyond conventional astrophysical features.

Classification: H / T

Priority: A

Status: PROPOSED

Primary Test

Compare:

$$[Y=f(X_{\rm astro})]$$

against

$$[Y=f(X_{\rm astro},X_{\rm arith})]$$

Primary Endpoint

$R^2_{\rm OOS,arith+astro}$

$R^2_{\rm OOS,astro}$.
]

Secondary metrics:

- RMSE;
- MAE;
- calibration;
- residual structure.

SFR-A002 — Arithmetic-Only SFR Prediction

Statement

Arithmetic-derived variables may contain predictive information about SFR independently of conventional astrophysical variables.

Classification: H / T

Status: PROPOSED

Required Comparison

$$[Y=f(X_{\rm arith})]$$

against:

[
 $Y=f(X_{\rm null})$
]

and

[
 $Y=f(X_{\rm astro})$.
]

This experiment is intentionally more demanding than incremental prediction.

SFR-E001 — Extreme-SFR Model Behavior

Historical S.T.A.R. models exhibited substantially different performance in extreme-SFR populations compared with the bulk sample.

Classification: E

Status: HISTORICAL OBSERVATION

Historical values include examples such as:

- full-sample RF ($R^2 \approx 0.8455$);
- full-sample GB ($R^2 \approx 0.8476$);
- non-outlier RF ($R^2 \approx 0.8729$);
- non-outlier GB ($R^2 \approx 0.8766$).

These are historical model results and must not be interpreted as evidence for the arithmetic hypothesis.

SFR-A003 — Extreme-SFR Robustness

Statement

Any proposed arithmetic/SFR relationship should be evaluated separately across predefined SFR strata rather than relying exclusively on aggregate performance.

Classification: T

Status: PROPOSED METHODOLOGICAL REQUIREMENT

17. Machine-Learning Controls

CTRL-RF001 — Random Feature Baseline

Statement

Randomly selected features can establish a baseline for predictive performance and help distinguish model capacity from theory-specific information.

Classification: E / M

Status: CONTROL

Historical `model15.py` work provides a starting point.

CTRL-A001 — Arithmetic Permutation Control

Statement

If arithmetic features contain meaningful information, permutation of their associations with astronomical objects should reduce the corresponding out-of-sample signal while preserving their marginal distributions.

Classification: T

Status: PROPOSED CONTROL

CTRL-A002 — Leakage-Controlled Reconstruction

All historical arithmetic/SFR models shall be reconstructed with:

- fold-local preprocessing;
- fold-local normalization;
- fold-local feature construction;
- fold-local redshift binning where applicable;
- no target leakage;

- independent test data.

Classification: T / M

Status: REQUIRED VALIDATION CONTROL

18. Data Provenance Claims

DATA-M001 — Coordinate Matching

Statement

A reproducible cross-survey catalog can be constructed using coordinate matching with a stated angular tolerance.

Historical S.T.A.R. matching used:

```
[  
  \theta_{\rm match}\leq 2"  
]
```

with an `arcsec_separation` diagnostic.

Classification: M / E

Status: INFRASTRUCTURE

The matching tolerance is a methodological parameter, not physical evidence.

DATA-M002 — Multi-Survey Integration

S.T.A.R. has assembled data from:

- SDSS;
- MaNGA;
- Pipe3D;
- HI;
- Galaxy Zoo;
- stellar-mass catalogs;

- environment measurements;
- photometry;
- 2MASS;
- UKIDSS;
- VizieR-derived catalogs.

Classification: E

Status: DATA INFRASTRUCTURE

The existence of these integrated datasets does not establish any theoretical claim.

19. RTCH Claims

RTCH-M001 — Variational RTCH Construction

Statement

The proposed RTCH action generates the stated field equations under the declared assumptions and variational conventions.

Classification: D / M

Status: DERIVATION AUDIT REQUIRED

The action includes candidate components:

```
[
S=
S_{\rm EH}
+S_{\phi}
+S_Y
+S_B
+S_{\rm top}
+S_m
+S_{\rm GHY}.
]
```

The complete derivation must be checked independently before treating the resulting equations as established.

RTCH-M002 — Closed Pullback Two-Form

Given

$$\left[\begin{array}{l} d_{\{\mathcal{T}\}}\omega=0 \end{array} \right]$$

and

$$\left[\begin{array}{l} F_Q=\Phi^*\omega, \end{array} \right]$$

the pullback satisfies

$$\left[\begin{array}{l} dF_Q=0. \end{array} \right]$$

Classification: D

Status: DERIVATION

The requirement that

$$\left[\begin{array}{l} [\omega]\neq 0 \end{array} \right]$$

distinguishes a closed non-exact topological sector from a trivial exact construction.

RTCH-H001 — Topological Coupling

Statement

A nontrivial topological coupling may produce observable deviations from standard gravitational/matter dynamics.

Classification: H / T

Priority: B

Status: PROPOSED

RTCH-H002 — Conformal Trace Coupling

For the candidate conformal coupling,

$$A^2(\phi, \mathcal{M}, \mathcal{J}_Q) g_{\mu\nu},$$

the matter coupling may generate an effective fifth force.

For a perfect fluid,

$$\bar{T} = -\epsilon + 3p.$$

For radiation,

$$p = \frac{\epsilon}{3}$$

and therefore

$$\bar{T} = 0.$$

Classification: D / T

Status: DERIVATION + PHYSICAL TEST REQUIRED

RTCH-H003 — Fifth-Force Acceleration

The point-particle action

$$S_p = -mc \int A, ds$$

produces candidate acceleration

$$-c^2h^{\mu\nu}\nabla_\nu\ln A.$$

Classification: D / T

Status: DERIVATION AUDIT / OBSERVATIONAL CONSTRAINT

RTCH-H004 — Unified Matter Description

Point-particle and perfect-fluid matter should be treated as different realizations of the same universal matter coupling rather than as two unrelated physical theories.

Classification: M

Status: THEORETICAL ARCHITECTURE

This does not require a literal action in which both matter descriptions simultaneously appear.

20. Symbolic Arithmetic Field

FIELD-001 — Arithmetic Topography

Statement

A scalar arithmetic field may be constructed as

$$f(l_1(E),\ldots,l_n(E))$$

with associated gradient

$$-\nabla h_{\rm arith}.$$

Classification: M / H

Status: CANDIDATE CONSTRUCTION

The gradient shall not be identified with gravity without independent physical validation.

FIELD-002 — Arithmetic Field as Gravity

Statement

Arithmetic gradients may contribute to gravitational or effective gravitational dynamics.

Classification: H / S

Status: UNVALIDATED

Required tests include comparison with:

- matter density;
- gravitational potential;
- lensing;
- peculiar velocity;
- galaxy dynamics;
- morphology;
- SFR.

FIELD-003 — Arithmetic Tensor / Einstein Equation Extension

A symbolic arithmetic tensor may provide an additional term in gravitational field equations.

Classification: H / S

Status: FUTURE THEORY

No claim that arithmetic replaces the Einstein tensor shall be treated as established.

21. Entropy/Cohomology Claims

ECC-001 — Entropy/Cohomology Structure

The ECC framework proposes that entropy-related quantities may be represented through a cohomological/topological construction involving closed but non-exact forms.

Classification: H / M

Status: THEORETICAL CANDIDATE

ECC-002 — Closed Non-Exact Requirement

A nontrivial topological construction requires a form satisfying

$$\begin{aligned} &[\\ &d\omega=0, \\ &\quad \quad \quad \\ &[\omega]\neq 0. \\ &] \end{aligned}$$

Classification: D / M

Status: DERIVATION / MODEL REQUIREMENT

Historical formulations relying on expressions equivalent to $(d(\text{d}\mathcal{M})=0)$ shall not be treated as sufficient evidence of nontrivial cohomology.

ECC-003 — Entropy–Energy Relation

The proposed entropy/cohomology structure may contribute to a generalized mass-energy relation.

Classification: H / T

Status: FUTURE EXPERIMENTAL/THEORETICAL CLAIM

Any candidate relation must satisfy:

1. dimensional consistency;
2. known-limit recovery;
3. conservation;
4. variational origin where claimed;
5. stability;
6. well-posed dynamics;

7. novel testable prediction.
-

22. Mass-Energy Generalization

ENERGY-001

A generalized thermodynamic/cohomological mass-energy relation may contain

$$\begin{bmatrix} E=mc^2 \end{bmatrix}$$

as an appropriate limiting case.

Classification: H / T

Status: FUTURE THEORY

The known relation must be recovered rather than merely numerically approximated.

23. Cosmological Expansion

COSMO-E001

Statement

Arithmetic/topological structure may evolve with redshift in a way that corresponds to cosmic evolution.

Classification: H / T

Status: PROPOSED

The experiment should compare:

$$\begin{bmatrix} \mathcal{R}(\tau) \end{bmatrix}$$

against independently derived cosmic evolution quantities:

[
 $D_{\rm cosmic}(z)$.
]

Persistent homology provides a candidate framework for comparing structure across redshift.

24. Hubble–Planck Tension

COSMO-E002

Statement

The S.T.A.R. framework may eventually provide a mechanism capable of explaining discrepancies between independent cosmological determinations of expansion parameters.

Classification: H / S

Status: DOWNSTREAM / FUTURE

This is not a design criterion for selecting the model.

The project must first establish the underlying arithmetic, topological, and dynamical correspondences.

25. Historical Claims Reclassified

The following historical claims are retained for provenance but are not accepted as established evidence:

HIST-001

“The universe is an arithmetic projection.”

Current Status: S / HISTORICAL

HIST-002

“Elliptic-curve rank directly determines cosmic structure.”

Current Status: H / HISTORICAL

HIST-003

“Rank 0 corresponds to voids.”

Current Status: H / T

HIST-004

“Rank 1 corresponds to filaments/galaxies.”

Current Status: H / T

HIST-005

“Rank 2 corresponds to clusters.”

Current Status: H / T

HIST-006

“Rank 3 corresponds to superclusters.”

Current Status: H / T

HIST-007

“Historical Wasserstein thresholds establish fidelity.”

Current Status: EXPLORATORY / HISTORICAL

HIST-008

“>98% correspondence/fidelity demonstrates the theory.”

Current Status: HISTORICAL MODEL RESULT

Such numerical results may be re-entered as evidence only after their data generation, normalization, null construction, independence, and statistical methodology have been reconstructed.

26. Machine-Learning Evidence Policy

Machine learning may be used for:

1. feature discovery;
2. nonlinear association detection;
3. predictive benchmarking;
4. residual analysis;
5. symbolic-regression hypothesis generation.

Machine learning shall not be treated as mathematical proof.

The required pipeline is:

```
[
\boxed{
\text{ML discovery}
\rightarrow
\text{candidate equation}
\rightarrow
\text{independent validation}
\rightarrow
\text{analytic derivation}
}
]
```

A high (R^2), low loss, SHAP importance, or symbolic-regression expression does not by itself establish a physical law.

27. Leakage and Contamination Policy

Historical S.T.A.R. scripts contain cases where derived quantities, normalization, redshift binning, feature construction, or related transformations may have been calculated before the train/test split.

These results shall therefore receive:

Leakage Status: POTENTIAL LEAKAGE

until reconstructed.

A historical model shall not be promoted to controlled evidence until:

$$\left[\begin{array}{l} \text{\texttt{\textbackslash varnothing}} \\ \end{array} \right]$$

at the appropriate object/field level and all learned preprocessing parameters are derived exclusively from the training data.

28. Statistical Evidence Policy

Statistical significance is not equivalent to practical predictive usefulness.

The registry therefore distinguishes:

Level 1 — Statistical Correspondence

A statistically significant relationship is detected.

Level 2 — Cross-Dataset Replication

The relationship survives independent datasets.

Level 3 — Predictive Correspondence

The relationship produces useful out-of-sample predictions.

Level 4 — Physical Theory

A mathematically coherent physical mechanism reproduces established limits and makes novel successful predictions.

Thus:

$$\left[\begin{array}{l} \text{\texttt{\textbackslash text\{significance\}}} \\ \text{\texttt{\textbackslash not\Rightarrow}} \\ \text{\texttt{\textbackslash text\{predictive utility\}}} \\ \text{\texttt{\textbackslash not\Rightarrow}} \\ \text{\texttt{\textbackslash text\{physical theory\}}}. \end{array} \right]$$

29. Null Model Requirements

Every empirical claim must specify an appropriate null.

Candidate null families include:

N1 — Permutation Null

Arithmetic assignments randomly permuted relative to astronomical objects.

N2 — Spatial Randomization

Spatial coordinates randomized while preserving relevant marginal distributions.

N3 — Invariant Replacement

Rank replaced by another arithmetic invariant.

N4 — Mapping Replacement

Candidate mapping replaced by a control mapping.

N5 — Synthetic Distribution Null

Data generated from known distributions without the proposed relationship.

N6 — Astrophysical Baseline

Conventional astrophysical variables used without arithmetic variables.

N7 — Random Feature Control

Randomly selected features used to establish model-capacity baselines.

30. Claim Promotion Ladder

A claim may progress through:

[
\boxed{

```

\text{Proposed}
\rightarrow
\text{Derived/Observed}
\rightarrow
\text{Computationally Reproduced}
\rightarrow
\text{Controlled}
\rightarrow
\text{Replicated}
\rightarrow
\text{Predictive}
\rightarrow
\text{Physical}
}
]

```

No claim may skip directly from exploratory observation to physical interpretation.

31. Priority A Claims

The following claims should receive immediate experimental attention:

ID	Claim
CORE-001	Arithmetic invariants may correspond to cosmic structure
RANK-001	Rank may encode cosmic environment
RANK-002	Rank may correspond to topological complexity
MAP-001	Candidate arithmetic projection can be tested
MAP-005	Alternative mappings can be compared
SFR-A001	Arithmetic features may add OOS SFR information
SFR-A002	Arithmetic-only SFR prediction may contain signal
SFR-A003	Arithmetic/SFR relation must survive population stratification
CTRL-A001	Arithmetic permutation provides a null
CTRL-A002	Leakage-controlled reconstruction is required

MATH-K001	($K_{\rm theory} \rightarrow 1$) requires formal reconstruction
COSMO-L001	($L_{\rm cosmo}$) requires baseline comparison

32. Priority B Claims

Priority B theoretical/physical work includes:

ID	Claim
RTCH-M001	RTCH action produces stated equations
RTCH-H001	Topological coupling may produce observable deviations
RTCH-H002	Trace coupling vanishes for radiation
RTCH-H003	Candidate fifth-force dynamics
FIELD-001	Arithmetic topography
FIELD-002	Arithmetic field contribution to gravity
ECC-001	Entropy/cohomology architecture
ECC-003	Generalized entropy-energy relation
ENERGY-001	($E=mc^2$) limiting relation
COSMO-E001	Redshift evolution of arithmetic topology
COSMO-E002	Possible downstream Hubble–Planck application

33. Required Experiments Generated by the Registry

The Claim Registry shall connect to the Experiment Registry through the following candidate experiment families.

EXP-K001

Log-normal Reciprocal-Normalization Test.

EXP-L001

($L_{\rm cosmo}(s)$) Model Validation.

EXP-SFR-A01

Arithmetic Incremental SFR Prediction.

EXP-SFR-A02

Arithmetic-Only SFR Prediction.

EXP-SFR-A03

Extreme-SFR Robustness.

EXP-CTRL-A01

Random-Feature Baseline.

EXP-CTRL-A02

Arithmetic Permutation Control.

EXP-RTCH-B01

RTCH Mathematical Consistency.

EXP-RTCH-B02

RTCH Fifth-Force Constraint.

EXP-RTCH-B03

RTCH Trace-Free Radiation Control.

Existing experiment identifiers shall take precedence if these identifiers have already been assigned. New experiments must be cross-referenced rather than duplicated.

34. Data Provenance Registry Requirement

Because the R&D audit identified extensive cross-survey data construction, the Claim Registry shall no longer operate independently of data provenance.

Each empirical claim must point to a dataset lineage:

```
[  
D_{\rm raw}  
\rightarrow  
D_{\rm matched}  
\rightarrow  
D_{\rm cleaned}  
\rightarrow  
D_{\rm analysis}  
\rightarrow  
D_{\rm model}  
\rightarrow  
D_{\rm test}.  
]
```

Every transformation must record:

- source dataset;
- destination dataset;
- matching rule;
- filtering;
- missing-value policy;
- duplicate policy;
- coordinate tolerance;
- derived columns;
- software version;
- date.

35. Historical R&D Preservation Policy

The R&D archives contain valuable historical development material.

The registry shall preserve:

- earlier models;
- scripts;
- unsuccessful experiments;
- contradictory results;
- unstable statistics;
- alternative mappings;
- discarded hypotheses;
- screenshots;
- intermediate datasets;
- previous parameter values.

However:

Historical preservation does not imply evidentiary endorsement.

Historical artifacts remain scientifically valuable because they document the development path and can reveal:

- reproducibility failures;
- hidden assumptions;
- parameter sensitivity;
- data-selection effects;
- successful ideas requiring formalization;
- negative evidence.

36. Registry Interpretation Rule

When a historical result conflicts with a later controlled result, both shall remain in the provenance record.

The later result shall not erase the earlier result.

Instead, the registry shall record:

```
[
\text{Historical Result}
\rightarrow
\text{Reanalysis}
\rightarrow
\text{Controlled Result}
]
```

and explain the reason for the change where known.

37. Current Scientific Posture

The S.T.A.R. program currently treats the following as the central open question:

```
[  
  \boxed{  
    \text{Can arithmetic invariants be mapped into geometric/topological structures in a way that  
    produces statistically nontrivial, reproducible correspondence with independent astronomical  
    data?}  
  }  
]
```

The project does not presently assume that:

- elliptic curves determine the universe;
- rank is physically fundamental;
- Fibonacci/Lucas sequences are uniquely relevant;
- $(z=200r)$ is physically correct;
- arithmetic gradients are gravity;
- arithmetic tensors replace Einstein's tensor;
- entropy/cohomology produces a new mass-energy law;
- RTCH describes nature;
- the Hubble–Planck tension is caused by arithmetic structure.

These remain hypotheses, model constructions, conjectures, or speculative extensions until the appropriate mathematical and empirical tests are completed.

38. Registry-to-Experiment Rule

Every active empirical claim must ultimately have a corresponding Experiment Registry entry containing:

1. Experiment ID;
2. Claim ID;
3. dataset;
4. dataset version;

5. null model;
6. competing baseline;
7. locked parameters;
8. train/test design;
9. preprocessing rules;
10. contamination controls;
11. primary endpoint;
12. secondary endpoints;
13. success criterion;
14. failure criterion;
15. replication requirement;
16. code artifact;
17. result artifact.

Therefore:

```
[  
\boxed{  
\text{No experiment without a claim}  
}  
]
```

and

```
[  
\boxed{  
\text{No empirical claim without a registered experiment}  
}  
]
```

unless the claim is explicitly classified as historical, observational, or infrastructural.

39. Registry-to-Results Rule

Experimental outputs shall not overwrite claims.

Instead:

```
[  
\text{Claim}  
\rightarrow  
\text{Experiment}]
```

```
\rightarrow  
\text{Result}  
\rightarrow  
\text{Evidence Update}.  
]
```

This preserves the distinction between a hypothesis and the result of testing that hypothesis.

40. Version Control

This document supersedes the structural organization of Claim/Evidence Registry v0.1.

Version 0.2 introduces:

1. explicit epistemic classification;
 2. explicit evidence provenance;
 3. explicit leakage/contamination status;
 4. historical-result preservation;
 5. negative-evidence recording;
 6. data-provenance linkage;
 7. experiment linkage;
 8. replication status;
 9. separate mathematical and empirical evidence;
 10. explicit ML evidence policy;
 11. explicit null-model requirements;
 12. separation of statistical significance from predictive utility;
 13. Priority A and Priority B claim groups;
 14. explicit reclassification of historical S.T.A.R. claims;
 15. integration of the R&D archive into the controlled research architecture.
-

41. Final Registry Principle

The Claim/Evidence Registry is not intended to prove the S.T.A.R. framework.

Its purpose is to make it possible to determine, claim by claim, **what has actually been established, what has merely been observed, what remains hypothetical, what has failed, and what experiment is required next.**

The registry therefore treats falsification, negative results, instability, and unresolved contradictions as first-class scientific information.

```
[
\boxed{
\text{Claim}
\neq
\text{Evidence}
\neq
\text{Experiment}
\neq
\text{Result}
\neq
\text{Theory}
}
]
```

Maintaining these distinctions is a prerequisite for the scientific development of S.T.A.R.